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Numerical simulation of equilibrium air plasma flow in the induction chamber of a high-power plasmatron

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Abstract Multi-parameter study on the subsonic equilibrium air plasma flows in the cylindrical discharge channel of an induction RF-plasmatron IPG-3 with the maximum operating power of 1 MW is presented. Simulations are carried out by numerical solution of full Navier–Stokes equations coupled with a two-dimensional equation for high-frequency electric field; the discretization of the equations is performed on a staggered Cartesian mesh. An effective method for calculation of the transport coefficients for ionized multi-component air (including the plasma conductivity) in the temperature range 300–15000 K is applied, with the higher-order approximations by the Sonine polynomial in the Chapman–Enskog method. Simulation results are presented, including the flow fields, electric field, and thermodynamic parameters of the air plasma in the discharge channel of a high-power induction plasmatron. The results obtained reveal for the first time the structure and parameters of a plasma torch in a large-diameter (200 mm) discharge channel at powers up to 300 kW, providing the axial specific enthalpies at the channel outlet as high as 68 MJ/kg.

Keywords RF-plasmatron · Dissociated air · Ionization · Subsonic flow · Heat transfer · High-frequency electric field

List of symbols

z	Axial coordinate
r	Radial coordinate
D_c	Diameter of the discharge channel
R_c	Radius of the discharge channel
L_c	Length of the discharge channel
ρ	Density of the gas/plasma mixture
$\mathbf{U} = (u, v, w)$	Velocity vector with axial, radial, and circumferential components
P	Pressure
μ	Viscosity of the gas/plasma mixture
h	Total enthalpy
λ_{eff}	Effective thermal conductivity
$c_{P, \text{eff}}$	Effective specific heat capacity at constant pressure
$\mathbf{F}$	Lorentz force (per unit volume of gas)
F_z	Lorentz force axial component
F_r	Lorentz force radial component

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$\dot{Q}_J$	Joule heat release rate (per unit volume of plasma)
$\mathbf{E}(z, r)$	Complex-valued amplitude of the electric field
$\mathbf{H}(z, r)$	Complex-valued amplitude of the magnetic field
$E_\theta = E_{V\theta} + E_{P\theta}$	Tangential component of the electric field amplitude
$E_{V\theta}$	Part of E_θ generated by the inductor in vacuum
$E_{P\theta}$	Part of E_θ generated by the Foucault's currents in plasma
E_z	Axial component of $\mathbf{E}(z, r)$
E_r	Radial component of $\mathbf{E}(z, r)$
j_θ	Tangential component of vortical current amplitude in plasma
ω	Circular frequency of the electric current in the coil
σ	Plasma electric conductivity
μ_0	Permeability of vacuum
T	Temperature of gas/plasma
T_w	Wall temperature
m	Molar mass of gas/plasma mixture
R	Universal gas constant
ξ	Number of terms in the Sonine polynomials expansions for transport coefficients in the Chapman–Enskog method
N_{ap}	Electric power consumed by the facility
N_{pl}	Power input to plasma
U_c	Axial velocity at the channel outlet
h_c	Specific enthalpy at the channel outlet
U_{c0}	Axial velocity at the center of the channel outlet
h_{c0}	Specific enthalpy at the center of the channel outlet
T_{c0}	Temperature at the center of the channel outlet
ψ	Non-dimensional stream function

1 Introduction

Plasma is a state of matter at high temperatures featured by significant dissociation of molecules into atoms, as well as by ionization leading to the appearance of charged particles, ions and free electrons. Due to the multitude of interactions between the neutral and charged particles, plasma is featured by complex thermodynamics, chemistry, and electromagnetic phenomena [1,2]. Definitely, plasma physics is one of the rapidly developing and prolific avenues of modern science, covering a wide spectrum of problems, from astrophysics to electric gas discharges, material science etc. [3]. Also, of significant fundamental and applied values are studies into plasma flow dynamics, where the presence of intrinsic or external electromagnetic fields plays an important role, as in magnetohydrodynamics (e.g., [4–6]).

The presence of charged particles in plasma makes its interactions with electric and magnetic fields quite complex and parameter dependent. In particular, plasma can be generated and sustained in a static electric field (arc or spark discharge), as well as in high-frequency fields (inductively coupled plasma) [2]. Several types of low- and high-frequency plasma waves are known, including those where electromagnetic field-charged particle interactions are coupled with pressure perturbations (so-called ion-acoustic waves) [7]. Such highly nonlinearly coupled interactions are observed not only in gases, but also in other media (e.g., porous materials saturated with liquid crystals [8]).

One of the scientific fields where plasma interaction with solid matter is of particular importance is the development of protective coatings for re-entry space vehicles, capable of withstanding the high heat fluxes and ablative wear of surfaces [5]. Experimental testing of thermal protection materials, studying the catalytic properties of materials and their heat endurance requires the development of appropriate plasma sources capable of providing high-enthalpy flows modeling the boundary layer conditions at different re-entry path altitudes. Currently, two main types of plasma sources, referred to as plasmatrone, are available. In the arc plasmatrone, the plasma torch is generated in a high-voltage arc discharge between anode and cathode, whereas in the induction coupled plasma generators (RF-plasmatrone) the torch is generated in the discharge channel by the high-frequency vortex electromagnetic field created by a coaxial coil. One of the advantages of the latter plasmatron type is that the plasma obtained is very clean, as it has no direct contact with the electrodes.

Several powerful induction coupled plasma generators are currently in operation worldwide. In Russia, such facilities are available at TSNIIMASH [9–11] (U-13 facility), TSAGI (VAT-104 facility) [12, 13], as well

as at IPMech RAS (IPG-3 plasmatron) [14]. The induction RF-plasmatrons IPG-3, and a smaller-scale IPG-4, operating at IPMech RAS are included in the List of unique research facilities of Russian Federation [15]. These facilities are used to solve a wide spectrum of fundamental and applied aerophysical problems, including the aerothermodynamics of plasma and high-enthalpy flows of molecular gases, heat transfer in equilibrium and non-equilibrium chemically reacting gas flows, testing the thermal protection materials for thermochemical resistance at re-entry conditions.

In the Western Europe, facilities of the same class are operational at von Karman Institute VKI (Belgium) [16–18], at the Institute of Space Systems IRS (Stuttgart University, Germany, PWK3 facility) [19,20], as well as at Japan Aerospace Exploration Agency (JAXA) [21–23]. These facilities have the following design and operation parameters: the power is up to 1.2 MW, diameter of the discharge channel is up to 200 mm, frequency of the generator is 400–2000 kHz, air flow enthalpy is up to 50 MJ/kg, and total pressure is in the range 0.001–1 atm.

In order to interpret correctly the results of experimental tests, it is vital to know the aerothermodynamical parameters at the outlet of the discharge channel of an RF-plasmatron. These parameters, in fact, determine the properties of subsonic and supersonic plasma jets in the test chamber where interaction with tested materials takes place. Few of these parameters are amenable to direct measurement; therefore, development and application of adequate numerical methods for simulation of multi-component chemically reacting plasma flows in the discharge channel is required. It is not coincidental, therefore, that many research groups operating the plasmatron facilities also develop and implement their numerical solvers. Numerical simulations of vortex flows of induction-generated plasma, performed by the coupled solution of Navier–Stokes equations and Maxwell equations for the HF electromagnetic field, were reported in [10,24–26]. The models used in these simulations have different account for non-equilibrium effects, geometry, computational schemes, etc.; the simulation results reflect the features of the particular facilities, as well as the operating parameters (pressure, power input, gas composition).

Simulations of the induction-generated plasma flows (air, nitrogen, carbon dioxide) in the discharge channel of IPG-4 facility (IPMech RAS) were carried out in [27,28], accompanying the experiments on the determination of the catalytic efficiency of high-temperature thermal protection coatings. For the cylindrical discharge channel of IPG-4 plasmatron, which is of 80 mm diameter and can have conical nozzles with the diameters 30, 40, and 50 mm, multi-parameter studies on the equilibrium and non-equilibrium multi-component air plasma flows were carried out [29,30].

The challenge in the numerical simulation of HF discharge in the molecular gas flow is to account for the tight coupling of the gas-dynamical and electrodynamical processes, multi-channel mode of energy transfer from the electric field to the plasma flow, taking into account the kinetic processes in the multi-temperature and multi-component plasma. In the numerical simulations of plasma flow in the discharge channel of the 100-kW HF plasmatron IPG-4, significant simplification of the electrodynamic part of the problem was applied. Namely, the equation describing the time-averaged complex-valued amplitude of the electric field in the circumferential direction was solved in the quasi-one-dimensional approximation [25].

However, it is questionable if such an approach can be applied to the modeling of the discharge channel of the IPG-3 plasmatron, having the diameter of 200 mm; also, IPG-3 facility operates at much higher power levels (up to 1 MW) and a different generator frequency. Therefore, the principal aim of the current work is the computational study on the subsonic air plasma flows in the discharge channel of the powerful HF plasmatron in wide ranges of operating conditions. The main focus of the research is on the comparison of one-dimensional and two-dimensional electric field models, as well as on the determination of plasma aerothermochemical parameters at the outlet of the discharge channel. Note that no similar studies have been performed so far for the 1 MW IPG-3 facility.

2 Problem statement

Axisymmetric flows of equilibrium air plasma in a cylindrical discharge channel of a powerful RF-plasmatron are considered. Gas ionization is achieved due to the high-frequency electromagnetic field generated by a coaxial external coil. Geometry of the discharge channel and the coil is sketched in Fig. 1. Air is supplied through a narrow annular inlet in the bottom end of the channel; the flow is swirling about the longitudinal axis for stabilization of the torch; also, flow swirl is required to avoid the contact between the hot plasma and channel wall.

The flow of equilibrium multi-component air plasma in the discharge channel is assumed to be steady-state and laminar. The rationale behind these assumptions is the following. The effects of plasma flow unsteadiness,

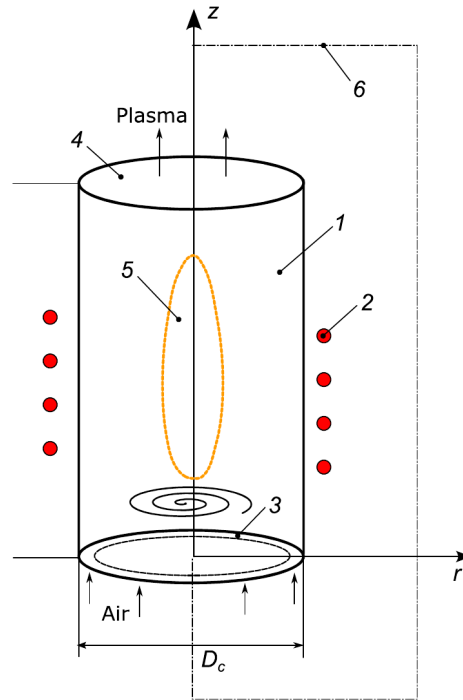


Fig. 1 Sketch of the discharge channel (1) with the induction coil (2), annular air inlet (3), plasma outlet (4), discharge zone (5) stabilized by flow swirl. Flow simulations are performed in the discharge channel (1), electric field is calculated in a wider domain shown by the dash-dotted line (6)

with nonlinear coupling between the plasma, electromagnetic field, and pressure waves, are very important at the transient stage of plasma torch formation, following the initial gas breakdown by the high-frequency electromagnetic field. However, the focus of the current research is on the plasmatron application to material testing where each test can last for minutes or dozens of minutes. The experiments show that once the torch is stabilized, it then remains quite stable, exhibiting no visible oscillations, or transient behavior. Also, experiments show that the flow in the discharge channel and in the test chamber is laminar (except maybe for the far field in the test chamber where some turbulization was observed on the plasma jet boundary). Therefore, the steady-state problem formulation is sufficient for the purpose of the current study, while unsteady formulation would be necessary in order to consider short-time transient behavior of plasma torch upon its ignition.

The flow of equilibrium plasma is described by the system of Navier–Stokes equations with the Lorentz force term in the momentum equation taking into account the effect of electromagnetic field on the charged particles. Also taken into account is the Joule heat released in plasma. The general form of the equations is well known in MHD studies [4,5]; the conservation laws are expressed as the continuity, momentum, and the energy equation formulated in terms of the total gas enthalpy:

$$\nabla(\rho \mathbf{U}) = 0 \quad (1)$$

$$\nabla(\rho \mathbf{U} \otimes \mathbf{U}) = -\nabla P + \nabla \boldsymbol{\tau} + \mathbf{F} \quad (2)$$

$$\nabla(\rho \mathbf{U} h) = \nabla \left(\frac{\lambda_{\text{eff}}}{c_{P,\text{eff}}} \nabla h \right) + \dot{Q}_J \quad (3)$$

Here, ρ is the density, $\mathbf{U} = (u, v, w)$ is the velocity vector with the axial, radial, and circumferential components, respectively; P is the pressure, $\boldsymbol{\tau} = \mu (\nabla \mathbf{U} + \nabla \mathbf{U}^T) - \frac{2}{3} \mu (\nabla \cdot \mathbf{U})$ is the viscous stress tensor, μ is the gas viscosity, h is the total gas enthalpy, λ_{eff} is the effective thermal conductivity, $c_{P,\text{eff}}$ is the effective specific heat capacity at constant pressure. Plasma radiation, as well as buoyancy effects, is not taken into account.

The problem is considered in the cylindrical coordinate system (z, r) shown in Fig. 1, and the coordinate form of Eqs. (1)–(3) can be found in [25]. The following boundary conditions are posed for the above equations. In the inlet, all required flow parameters are set, including the gas parameters (pressure, temperature, composition), the axial velocity component u (determining the gas flowrate) and circumferential velocity component w (determining the flow swirl); the radial velocity component in the inlet is $v = 0$. On the solid surfaces

(channel side and end walls), no-slip conditions are posed for the velocity; also, the prescribed temperature (or adiabatic conditions) is specified. On the fixed-pressure outlet, the known “soft” boundary conditions are posed to allow free outflow of plasma from the discharge channel. Finally, symmetry conditions are posed on the channel axis.

The source terms entering Eqs. (2) and (3) describe the interaction of plasma with the electromagnetic field generated by the coil. Namely, $\mathbf{F}$ is the Lorentz force acting on charged particles, while $\dot{Q}_J$ is the Joule heat release rate (both quantities are taken per unit volume of gas).

The electromagnetic field in the discharge channel is calculated under the following main assumptions: the plasma is quasi-neutral, the magnetic permeability of the plasma is $\mu = 1$, its dielectric permittivity, ε , is independent of the electric field, and hence, it does not depend on z and r . Following [24, 31], the electric and magnetic fields generated by the high-frequency current in the coil are assumed to be monochromatic: $\mathbf{E}(t, z, r) = \mathbf{E}(z, r) \exp(-i\omega t)$, $\mathbf{H}(t, z, r) = \mathbf{H}(z, r) \exp(-i\omega t)$, where ω is the circular frequency of the electric current in the coil, $\mathbf{E}(z, r)$ and $\mathbf{H}(z, r)$ are the complex-valued amplitudes of the fields, respectively. These assumptions are common in the field of CFD simulations for induction plasmatrons used in material testing, with experiments duration from several minutes to several hours, so that stable values of all the regime parameters are maintained during the long experiment time. Studies on the transient electromagnetic behavior in induction plasmatrons are presented in [32, 33].

In the radio frequency range, one can neglect the displacement current due to the conditions $\omega \ll \nu_e$ and $|\varepsilon| \ll 4\pi\sigma/\omega$ ($P = 0.01 - 1$ atm, $T \leq 15000$ K), where ν_e is the electron collision frequency and σ is the plasma electric conductivity [34].

In order to apply the axisymmetric problem formulation, the actual inductor coil turns are represented by the corresponding number of infinitely thin circular currents.

Under these assumptions, the Maxwell equations, written together with the Ohm's law $j_\theta = \sigma E_\theta$, are reduced to an elliptic-type equation for the tangential complex-valued amplitude of the electric field E_θ (the axial and radial components vanish due to the axial symmetry of the problem, $E_z = E_r = 0$):

$$\frac{\partial^2 E_\theta}{\partial z^2} + \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} r E_\theta \right) = -i\omega\mu_0\sigma E_\theta \quad (4)$$

Here, j_θ is the tangential component of the vortical current in plasma and $\mu_0 = 1.2566 \cdot 10^{-6}$ N/m is the permeability of vacuum. This complex-valued equation is solved with the symmetry boundary condition $E_\theta(z, 0) = 0$ on the axis; also, the amplitude of the electric field vanishes at infinity, $E_\theta(z, r) = 0$ for $r^2 + z^2 \rightarrow \infty$.

Following [24], we decompose the electric field into two parts, $E_\theta = E_{V\theta} + E_{P\theta}$, where $E_{V\theta}$ is the field generated by the inductor in vacuum, whereas $E_{P\theta}$ is the field generated by the Foucault's currents in the plasma. The amplitude $E_{V\theta}$ is conveniently expressed by an analytical formula for the electric field generated in vacuum by a high-frequency current running in an infinitely thin round loop conductor [35]. For the plasma-generated electric field, $E_{P\theta}$, the following equation holds:

$$\frac{\partial^2 E_{P\theta}}{\partial z^2} + \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} r E_{P\theta} \right) = -i\omega\mu_0\sigma (E_{V\theta} + E_{P\theta}) \quad (5)$$

The boundary conditions for $E_{P\theta}$ are the same as for the total field: $E_{P\theta}(z, 0) = 0$, $E_{P\theta}(z, r) = 0$ for $r^2 + z^2 \rightarrow \infty$. Numerical solution of Eq. (5) is easier than integration of Eq. (4) because the singularities arising due to the assumption of infinitely thin currents in the coil are avoided (the conductivity σ is zero outside the channel where the coil is located). Also, the amplitude of plasma-generated electric field $E_{P\theta}$ decays with the distance much faster than that of the induction coil $E_{V\theta}$, which facilitates the numerical approximation of boundary condition at infinity.

The source terms in Eqs. (2) and (3) are expressed in terms of the plasma conductivity and electric field amplitude and frequency in the following way. The axial, F_z , and radial, F_r , components of the Lorentz force are

$$F_z = -\frac{\sigma}{2\omega} \operatorname{Re} \left\{ E_\theta \left(i \frac{\partial E_\theta}{\partial z} \right)^* \right\}, \quad F_r = -\frac{\sigma}{2\omega} \frac{1}{r} \operatorname{Re} \left\{ E_\theta \left(i \frac{\partial r E_\theta}{\partial r} \right)^* \right\} \quad (6)$$

where Re denotes the real part of a complex-valued variable and the asterisk denotes the complex conjugate of the complex-valued quantity in the curly braces.

The Joule heating power per unit volume entering the energy equation (3) is given by

$$\dot{Q}_J = \frac{\sigma}{2} E_\theta E_\theta^* \quad (7)$$

Thus, the problem is reduced to the solution of the set of Eqs. (1)–(3), (5)–(7) describing the inductively coupled plasma flow. To close this system of equations, the ideal gas equation of state is used:

$$\rho = \frac{Pm}{RT} \quad (8)$$

where T is the temperature, R is the universal gas constant, and m is the mean molar mass of the gas. Also, thermodynamic, transport, and electric properties of plasma must be specified. The approach toward the calculation of plasma properties is described in the following section.

3 Equilibrium air plasma properties

In the framework of equilibrium air plasma model adopted in this work, the chemical composition, ionization, and transport properties (viscosity, heat conductivity, electric conductivity) of the gas were pre-calculated as functions of pressure and temperature in the ranges $0.01 \leq P \leq 1$ atm, $300 \leq T \leq 15000$ K, respectively. The effects of electromagnetic field on the transport properties were neglected. Detailed description of the approaches used can be found in [36–40]; here, a brief overview of the models is given, and examples demonstrating their validity are presented.

Chemical composition and thermodynamic properties of equilibrium gas mixtures were carried out by the SoVA software package developed by the authors. A system of nonlinear algebraic equations describing the chemical and ionization equilibrium was solved iteratively by the method [37]; no allowance for gas and plasma non-ideality was made.

The transport coefficients were calculated by the exact formulas of the Chapman–Enskog method [36–40]. Solutions were obtained as expansions in Sonine polynomials of different order ξ , providing the viscosity $\mu(\xi)$, heat conductivity due to translational degrees of freedom $\lambda_{tr}(\xi)$, due to the internal energy λ_{in} , and due to the equilibrium chemical reactions $\lambda_R(\xi)$, with the effective heat conductivity $\lambda_{eff}(\xi) = \lambda_{tr}(\xi) + \lambda_{in} + \lambda_R(\xi)$; the electric conductivity $\sigma(\xi)$ was also found. For neutral (non-ionized) species, acceptable accuracy is given by the first nonzero approximations (for viscosity $\xi = 1$, while for heat conductivity $\xi = 2$). Higher-order approximations are required, however, at high temperatures where ionization becomes significant. Therefore, calculations of viscosity were carried out for $\xi = 2$, and the third nonzero approximation was used for other transport coefficients ($\xi = 4$ for $\lambda_{tr}(\xi)$, and $\xi = 3$ for $\sigma(\xi)$). This approach provided the accuracy within 5%.

To demonstrate the validity of the current approach, in Fig. 2 the effective heat conductivity $\lambda_{eff}(\xi)$ and electric conductivity $\sigma(\xi)$ calculated by formulas [37–40] are plotted for partially ionized air at $P = 1$ atm. Calculations were performed in the third nonzero approximation, as well as for the first nonzero approximation. Also shown are the calculation data from work by Capitelli et al [41]. It can be seen that the current results for $\sigma(\xi = 3)$ practically coincide with the calculation [41], whereas the first approximation $\xi = 1$ underestimates the conductivity significantly at $T = 10000 - 15000$ K. For the effective heat conductivity $\lambda_{eff}(\xi)$, the results with $\xi = 4$ practically coincide with [41] at $T < 12000$ K, whereas at higher temperatures the current approach gives the conductivities which are higher by up to 15% than those from [41]. This difference can be attributed to the differences in the formulas for the components of λ_{eff} , as well as to the differences in the interaction potentials and collision integrals for the ionized air adopted in the current work and in [41]. The current calculation of λ_{eff} with $\xi = 2$ gives lower values than those obtained for $\xi = 4$ at $T = 10000 - 15000$ K. Interestingly, the final value of λ_{eff} for $\xi = 2$ practically coincides at $T = 15000$ K with the data [41]; however, this agreement is just occasional because in [41] calculations were performed in the higher-order approximations of the Chapman–Enskog method.

4 Numerical procedure

Numerical solution of two-dimensional Navier–Stokes equations in the axisymmetric framework was performed by the finite-volume method on a non-uniform Cartesian mesh. The pressure–velocity coupling was achieved by the known SIMPLE method [42] on a staggered grid. The convective terms were approximated by

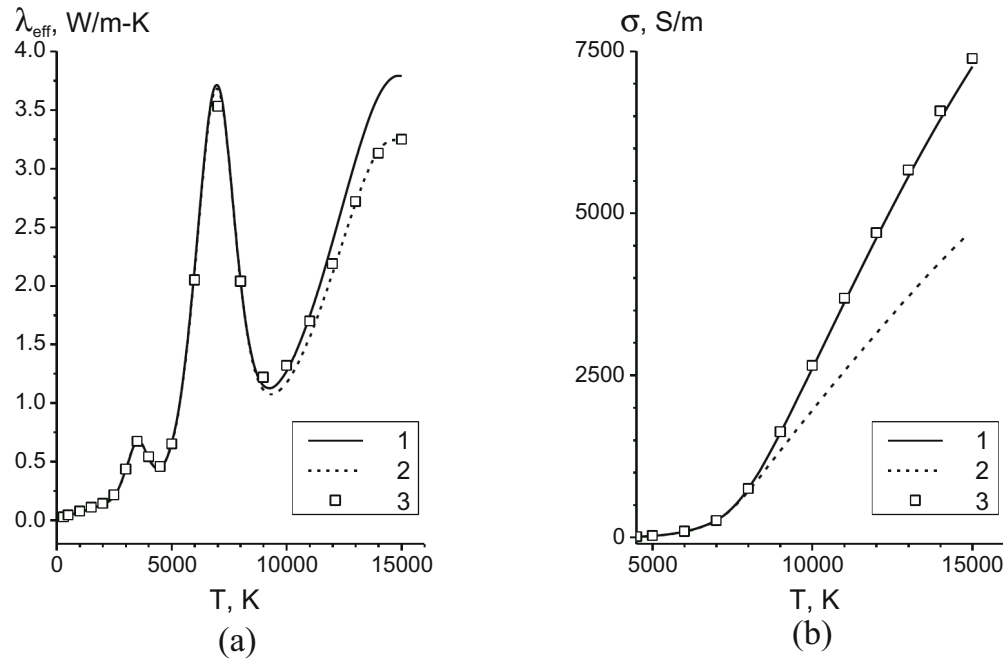


Fig. 2 Comparison of SoVA results with calculations [41] for partially ionized air at $P = 1$ atm: **a** effective heat conductivity $\lambda_{\text{eff}}(\xi)$ (1: $\xi = 4$, 2: $\xi = 2$) **b** electric conductivity $\sigma(\xi)$ (1: $\xi = 3$, 2: $\xi = 1$). Data [41] are shown by symbols 3

the backward differences, and the diffusion-type terms were approximated by the standard central-difference scheme. To enhance the solution stability during iterations, underrelaxation procedure was applied.

A numerical method for the solution of the elliptic equation for the complex-valued electric field amplitude (5) was developed earlier in [26], where numerical simulations of air plasma flow in the discharge channel of IPG-3 plasmatron were presented for the first time. The complex-valued equation (5) is presented as two coupled elliptic equations for the real and imaginary parts of the amplitude, approximated on a Cartesian mesh covering a larger area than the discharge channel (see the dash-dotted line in Fig. 1). To avoid the necessity to interpolate the electric field solution between different meshed, the mesh for solving (5) was generated by extending the mesh used for flow simulations in the radial and axial directions. In all the simulations presented hereafter, the radial size of the computational domain for the electric field was equal to $3R_c$, and its axial length was $1.5L_c$ (where $R_c = D_c/2$ and L_c are the radius and length of the discharge channel).

The system of algebraic equations resulting from the finite-volume approximation of Eq. (5) was solved by the effective iterative method GMRES with ILUT preconditioner [43].

5 Numerical results

5.1 Parameters

The computational model described in this work was applied to the numerical simulations of the equilibrium air plasma flow in the discharge channel of induction plasmatron IPG-3. Calculations were performed for the simple cylindrical shape of the discharge channel having the length $L_c = 870$ mm and diameter $D_c = 192$ mm. The distance from the inlet to the first turn of the inductor coil was 350 mm, the length of the four-turn inductor was 255 mm, and the diameter of the coil was 215 mm. The frequency of the alternating current in the inductor was 0.44 MHz. The efficiency of plasmatron, defined as the ratio of the power input to plasma, N_{pl} , and the electric power consumed by the facility, N_{ap} , was equal 0.6. In the numerical simulations, the input value was N_{pl} ; the power input to plasma was then calculated as $N_{\text{pl}} = 0.6N_{\text{ap}}$.

The electric field $E_{V\theta}$ in Eq. (5) is proportional to the amplitude of the current in the coil, however, the current itself is not a convenient input parameter: the principal quantity characterizing the power input to plasma is N_{pl} . In order to guarantee the correct power input, the current in the coils was not set constant, but was adjusted iteratively: on each iteration, the volume integral of the Joule heat release rate (7) was evaluated

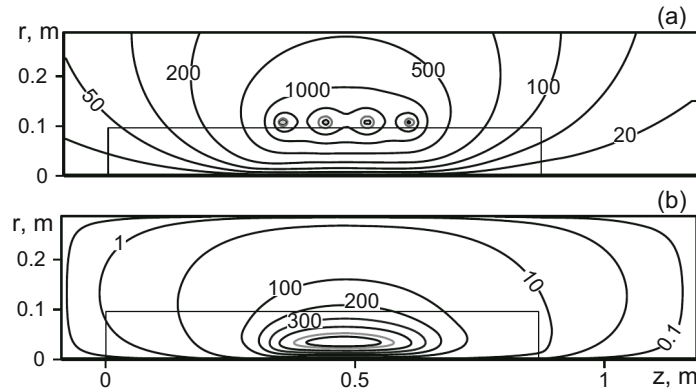


Fig. 3 Contours of the electric field amplitude E_θ **a** and $E_{p\theta}$ **b**, V/m, in the computational domain about the discharge channel of IPG-3 plasmatron. Simulation parameters are: $P = 100$ mbar, electric power $N_{ap} = 300$ kW, power input to plasma $N_{pl} = 180$ kW

and compared with the set value of N_{pl} ; the current was then increased or decreased in such a way that the power integral matched the required power input N_{pl} .

In all the simulations performed in the current work, the mass flowrate of air was fixed at $G = 11$ g/s. Air was supplied at the inlet plane $z = 0$ through a narrow annular slot of 3 mm width near the channel wall, the flow swirl angle was 45 degrees (i.e., the ratio of tangential and longitudinal velocity components was equal 1.0); the radial inlet velocity was zero. The temperature of the discharge channel walls was set at the room level, $T_w = 300$ K.

The numerical grid size in the discharge channel was 103×43 (in the axial and radial directions), the mesh was non-uniform (refined toward the axis in the radial direction, and toward the plasma region in the axial direction). The electric field was calculated on a 279×143 mesh.

5.2 Electromagnetic field

Consider now the electric field patterns obtained in the simulation of IPG-3 plasmatron at the pressure $P = 100$ mbar and electric power $N_{ap} = 300$ kW (therefore, the power input to plasma was $N_{pl} = 180$ kW).

In Fig. 3, the calculated amplitudes of electric field E_θ (a) and $E_{p\theta}$ (b) are plotted. The boundary of the discharge channel is indicated by a rectangle of the corresponding size. In Fig. 3 (a), local maxima of the electric field E_θ are clearly visible, indicating the positions of the inductor coil turns. The electric field component $E_{V\theta}$ is singular, and it tends to infinity on approaching the circular current in the coil turn; however, it does not affect the solution because there is no plasma beyond the channel, and the conductivity there is $\sigma = 0$. On the contour lines of Fig. 3 (a), the local maxima of E_θ are finite, though, because they are not coinciding exactly with any of mesh node. In the discharge channel, the maximum values of E_θ and $E_{p\theta}$ are equal to 2000 and 600 V/m, respectively.

5.3 Air plasma flow

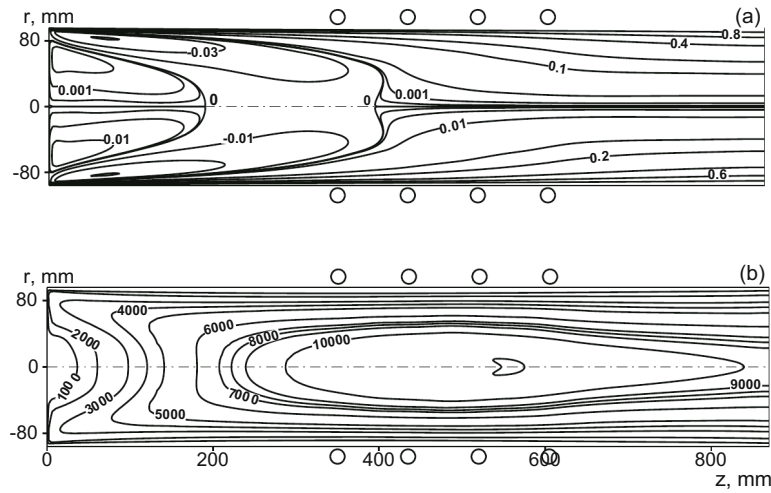
In Table 1, the case matrix is presented, and the main plasma parameters obtained in the channel outlet are listed. The first column is the IPG-3 operating mode number. The following notation is adopted for the axial values at the center of the channel outlet: U_{c0} is the longitudinal velocity, m/s; h_{c0} is the specific enthalpy, MJ/kg; and T_{c0} is the temperature at the center of the outlet section, K.

To demonstrate the flowfield developing in the discharge channel, in Fig. 4a,b the contour lines of the non-dimensional stream function ψ and isotherms are presented for the air plasma flow; these results were obtained in IPG-3 simulations with $P = 50$ mbar, $N_{ap} = 200$ kW, $N_{pl} = 120$ kW.

The stream function was obtained by integrating the longitudinal velocity component with respect to the radial coordinate between the axis (where the stream function was taken to be zero) up to the side wall of the channel. Taking into account the variable gas density and cylindrical geometry, the stream function is defined

Table 1 IPG-3 operating modes and main plasma parameters at the channel outlet

No.	P , mbar	N_{ap} , kW	N_{pl} , kW	U_{c0} , m/s	h_{c0} , MJ/kg	T_{c0} , K
1	50	50	30	11.5	16.3	5380
2	50	100	60	84.8	36.6	6930
3	50	200	120	261	57.0	9900
4	50	300	180	366	68.7	10630
5	100	50	30	6.3	15.7	5530
6	100	100	60	44.0	35.8	6950
7	100	200	120	145	52.7	9900
8	100	300	180	212	61.2	10620
9	200	50	30	3.7	16.3	5690
10	200	100	60	23.9	35.1	7230
11	200	200	120	80.6	48.0	9470
12	200	300	180	125	55.4	10390

**Fig. 4** Contours of non-dimensional stream function ψ **a** and temperature T , K **b** in the IPG-3 discharge channel at $P = 50$ mbar, $N_{pl} = 120$ kW (case 3 in Table 1)

as

$$\Psi(z, r) = \int_0^r \rho u r dr$$

The non-dimensional stream function ψ presented in Fig. 4,a is obtained by normalizing Ψ by its value on the channel wall. It can be seen that multiple vortices are developing in the channel near the inlet, whereas in its outlet part the streamlines are straightening and become parallel, indicating the nearly translational plasma flow. The maximum temperature attained in the channel in the induction zone is $T_{\max} = 10470$ K.

In Figs. 5, 6, and 7, the radial profiles of the enthalpy, h_c , and longitudinal velocity, U_c , at the discharge channel outlet are plotted for the twelve operating modes of the plasmatron listed in Table 1. Figure 5 corresponds to pressure $P = 50$ mbar, and Figs. 6 and 7 present the results obtained for pressure 100, and 200 mbar, respectively. In each of these figures, radial profiles are plotted for the power $N_{ap} = 50, 100, 200$, and 300 kW; the curve numbers correspond to the operational mode number in Table 1. The accuracy of the results for high-power operation modes with $N_{ap} = 300$ kW can be lower because in this case the enthalpy and temperature are too high for plasma radiation to be neglected [44,45].

Note that the low-power operation mode, $N_{ap} = 50$ kW, is featured by non-monotonic behavior of outlet velocity U_c : at the jet axis the velocity is low, then it increases toward the periphery and reaches its maximum and then decays rapidly toward the channel wall. The flow velocity, as well as the maximum enthalpy, decreases with the increase in the pressure because the gas density is increasing, while the mass flowrate is maintained constant.

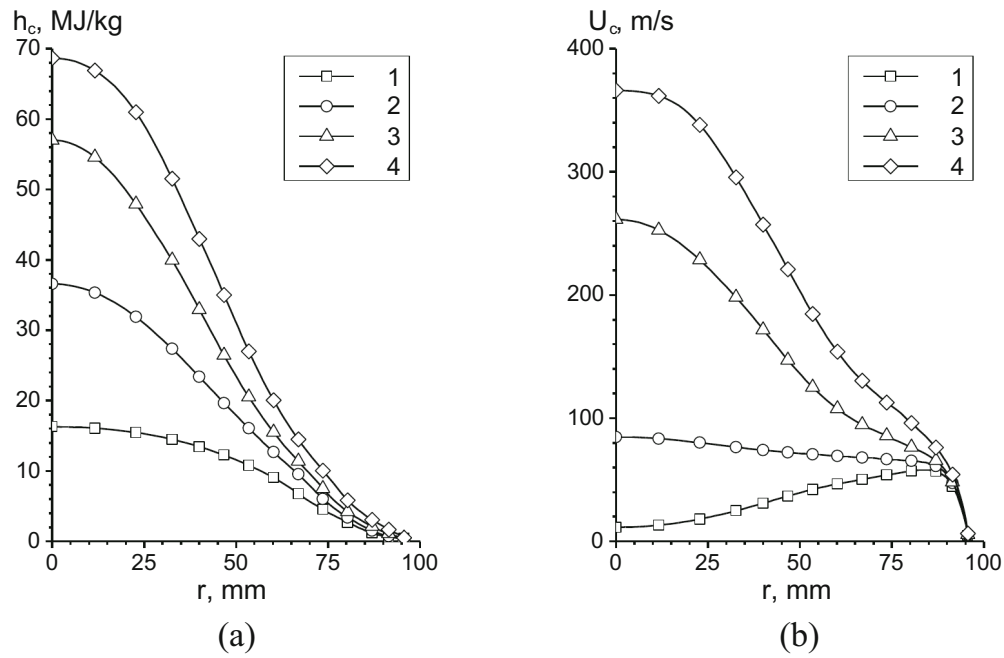


Fig. 5 Radial profiles of outlet enthalpy h_c **a** and longitudinal velocity U_c **b** for pressure $P = 50$ mbar: the curve numbers correspond to operating modes of IPG-3 plasmatron (see Table 1)

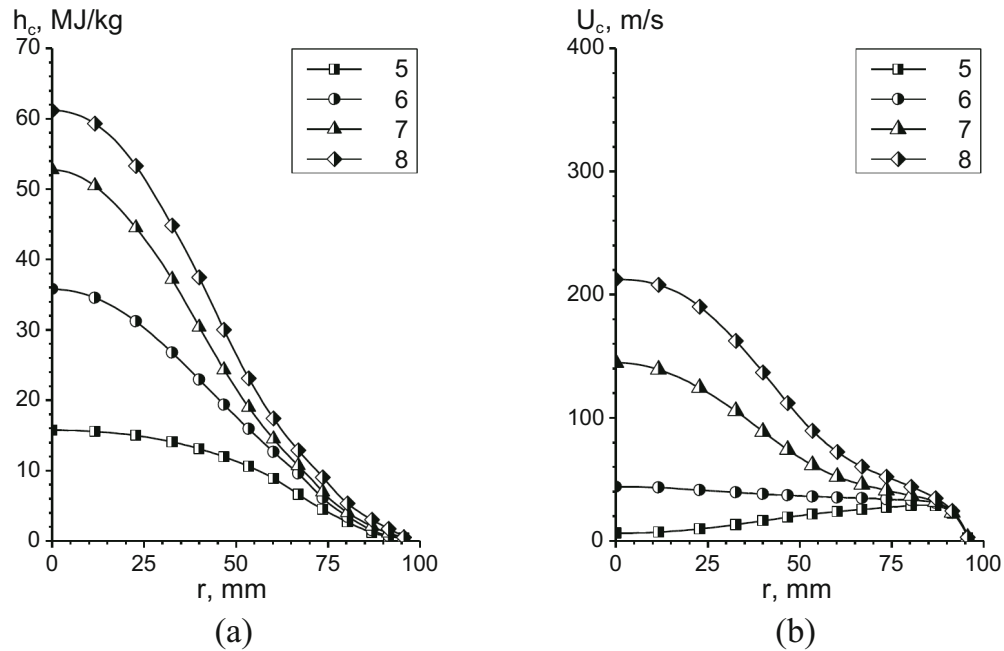


Fig. 6 Radial profiles of outlet enthalpy h_c **a** and longitudinal velocity U_c **b** for pressure $P = 100$ mbar: the curve numbers correspond to operating modes of IPG-3 plasmatron (see Table 1)

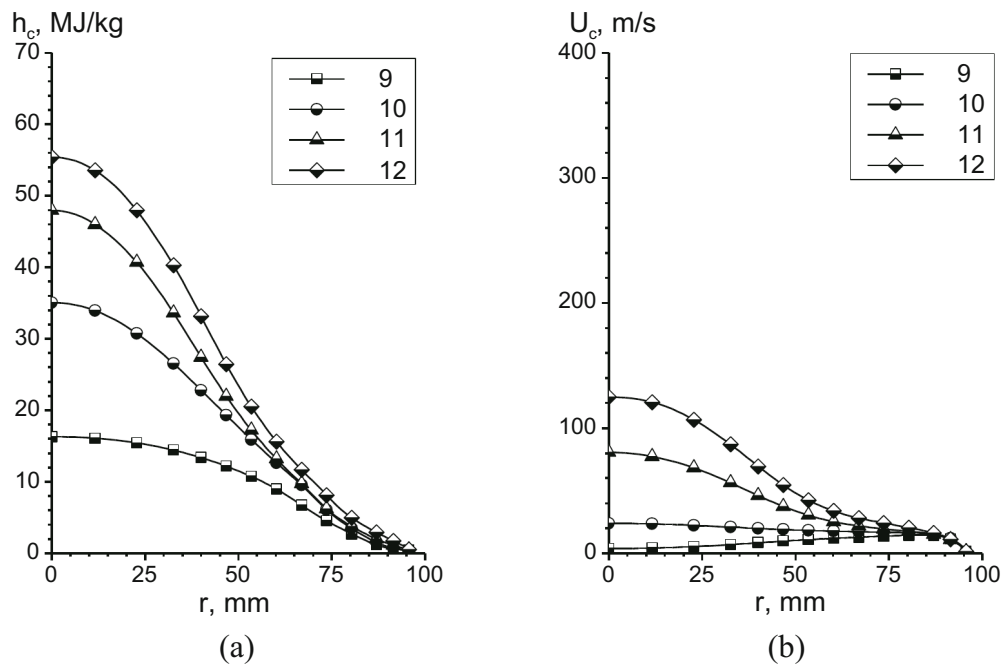


Fig. 7 Radial profiles of outlet enthalpy h_c **a** and longitudinal velocity U_c **b** for pressure $P = 200$ mbar: the curve numbers correspond to operating modes of IPG-3 plasmatron (see Table 1)

Conclusions

The computational model presented in this work is based on the coupled solution of Navier–Stokes equations for equilibrium plasma and two-dimensional Maxwell equation for the amplitude of high-frequency electromagnetic field. The model allows simulations of inductively coupled air plasma flows in the discharge channel in the whole range of operational parameters of a powerful IPG-3 plasmatron.

In the numerical simulations carried out in this work, the flow structure and plasma torch parameters were predicted for a large-size (200 mm diameter) discharge channel of IPG-3 plasmatron, extending significantly the power range (up to 300 kW) with respect to the previous simulations of a 80 mm discharge channel (IPG-4 facility).

Numerical calculations indicate the development of rather complex flowfield structure in the discharge channel of IPG-3 facility, with large primary vortex extending from the inlet to the inductor, and smaller secondary vortices near the inlet end of the channel, developing at high enough power input to plasma. It is obtained that, in the pressure and generator power ranges considered, air plasma flows with velocities of 44–260 m/s and enthalpies of 16.3–68.7 MJ/kg at the outlet center can be achieved. The predicted profiles of thermodynamic parameters at the discharge channel outlet give the possibility to model the aerodynamic heating in the experiments on high-frequency IPG-3 plasmatron.

The models developed and results obtained are valid under the assumption on thermal and chemical equilibrium in the plasma. For air plasma, this assumption is applicable at pressures $P \geq 0.05$ atm. At lower pressures, non-equilibrium effects may become significant, which requires further development of the model and software, inclusion of finite-rate chemistry, temperature non-equilibrium, and plasma radiation.

Note finally that in the present work, the mathematical model was based on the approach traditional for the reactive gas and plasma flow simulations. The solutions obtained indicate that the flowfield in the discharge channel is featured by high-gradient zones (see the stream function distribution in Fig. 4) developing due to the formation of recirculation zones in the swirling plasma flow. Also, large gradients are present in the high-frequency electric field in the conductive plasma due to the skin effect. Under such conditions, improved description of physical phenomena with multiple time and length scales requires more detailed consideration of the continua involved. In particular, it would be an interesting future development to apply a generalized continuum model, for example, the second-gradient continua approach, a modern concept that proved successful in many areas, including mechanics of solids, fluid dynamics and interphase phenomena [46,47].

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Declarations

Conflict of interest Authors state the absence of any conflict of interests or competing interests with respect to this paper and work conducted.

Data availability Not applicable.

Code availability Not applicable.

Author's contribution All authors contributed to the study conception and design. Problem statement was performed by AFK. Code development, model implementation and numerical simulations were performed by SAV and AIB. Analysis and generalization of results was performed by AFK, SAV, and SEY. All authors contributed to writing the manuscript. All authors read and approved the final manuscript.

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